Topic: Variation of parameters for nonhomogeneous systems

Question: Given the vector F and the complementary solution for a system of differential equations $\vec{x}' = A\vec{x} + F$, use variation of parameters to find the system's particular solution.

$$\vec{x_c} = c_1 \begin{bmatrix} 1 \\ -2 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 3 \\ 5 \end{bmatrix} e^{-3t}$$

$$F(t) = \begin{bmatrix} e^t \\ -\sin t \end{bmatrix}$$

Answer choices:

A $$\vec{x_p} = \begin{bmatrix} 7 \\ -25 \end{bmatrix} e^t + \frac{17}{47} \begin{bmatrix} \frac{41}{5} \\ 87 \end{bmatrix} \cos t - \frac{17}{47} \begin{bmatrix} \frac{623}{5} \\ 61 \end{bmatrix} \sin t$$

B $$\vec{x_p} = \begin{bmatrix} 7 \\ -25 \end{bmatrix} e^t - \frac{17}{47} \begin{bmatrix} \frac{623}{5} \\ 61 \end{bmatrix} \sin t$$

C $$\vec{x_p} = \frac{1}{22} \begin{bmatrix} -7 \\ 25 \end{bmatrix} e^t + \frac{1}{110} \begin{bmatrix} -3 \\ 17 \end{bmatrix} \cos t + \frac{1}{110} \begin{bmatrix} -15 \\ 9 \end{bmatrix} \sin t$$

D $$\vec{x_p} = -\frac{1}{22} \begin{bmatrix} 7 \\ -25 \end{bmatrix} e^t - \frac{11}{940} \begin{bmatrix} \frac{623}{5} \\ 61 \end{bmatrix} \sin t$$

© Krista King Math

Solution: C

We already have the complementary solution, so we'll use the solution vectors $\overrightarrow{x_1}$ and $\overrightarrow{x_2}$ to form $\Phi(t)$.

$$\Phi(t) = \begin{bmatrix} e^{2t} & 3e^{-3t} \\ -2e^{2t} & 5e^{-3t} \end{bmatrix}$$

Find the inverse $\Phi^{-1}(t)$ by changing $[\Phi(t)\,|\,I]$ into $[I\,|\,\Phi^{-1}(t)]$.

$$\left[\begin{array}{cc|cc} e^{2t} & 3e^{-3t} & 1 & 0 \\ -2e^{2t} & 5e^{-3t} & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & 3e^{-5t} & e^{-2t} & 0 \\ -2e^{2t} & 5e^{-3t} & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & 3e^{-5t} & e^{-2t} & 0 \\ 0 & 11e^{-3t} & 2 & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & 3e^{-5t} & e^{-2t} & 0 \\ 0 & 1 & \frac{2}{11}e^{3t} & \frac{1}{11}e^{3t} \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & 0 & \frac{5}{11}e^{-2t} & -\frac{3}{11}e^{-2t} \\ 0 & 1 & \frac{2}{11}e^{3t} & \frac{1}{11}e^{3t} \end{array}\right]$$

Now with

$$\Phi(t) = \begin{bmatrix} e^{2t} & 3e^{-3t} \\ -2e^{2t} & 5e^{-3t} \end{bmatrix} \qquad \Phi^{-1}(t) = \begin{bmatrix} \frac{5}{11}e^{-2t} & -\frac{3}{11}e^{-2t} \\ \frac{2}{11}e^{3t} & \frac{1}{11}e^{3t} \end{bmatrix} \qquad F(t) = \begin{bmatrix} e^{t} \\ -\sin t \end{bmatrix}$$

© Krista King Math

the particular solution from the method of variation of parameters is given by

$$\overrightarrow{x_p} = \Phi(t) \int \Phi^{-1}(t) F(t)\, dt$$

$$\overrightarrow{x_p} = \begin{bmatrix} e^{2t} & 3e^{-3t} \\ -2e^{2t} & 5e^{-3t} \end{bmatrix} \int \begin{bmatrix} \frac{5}{11}e^{-2t} & -\frac{3}{11}e^{-2t} \\ \frac{2}{11}e^{3t} & \frac{1}{11}e^{3t} \end{bmatrix} \begin{bmatrix} e^{t} \\ -\sin t \end{bmatrix} dt$$

$$\overrightarrow{x_p} = \begin{bmatrix} e^{2t} & 3e^{-3t} \\ -2e^{2t} & 5e^{-3t} \end{bmatrix} \int \begin{bmatrix} \frac{5}{11}e^{-2t}(e^{t}) - \frac{3}{11}e^{-2t}(-\sin t) \\ \frac{2}{11}e^{3t}(e^{t}) + \frac{1}{11}e^{3t}(-\sin t) \end{bmatrix} dt$$

$$\overrightarrow{x_p} = \begin{bmatrix} e^{2t} & 3e^{-3t} \\ -2e^{2t} & 5e^{-3t} \end{bmatrix} \int \begin{bmatrix} \frac{5}{11}e^{-t} + \frac{3}{11}e^{-2t}\sin t \\ \frac{2}{11}e^{4t} - \frac{1}{11}e^{3t}\sin t \end{bmatrix} dt$$

Integrate, then simplify the result.

$$\overrightarrow{x_p} = \begin{bmatrix} e^{2t} & 3e^{-3t} \\ -2e^{2t} & 5e^{-3t} \end{bmatrix} \begin{bmatrix} -\frac{5}{11}e^{-t} - \frac{3}{55}e^{-2t}(\cos t + 2\sin t) \\ \frac{1}{22}e^{4t} + \frac{1}{110}e^{3t}(\cos t - 3\sin t) \end{bmatrix}$$

$$\overrightarrow{x_p} = \begin{bmatrix} e^{2t}(-\frac{5}{11}e^{-t} - \frac{3}{55}e^{-2t}(\cos t + 2\sin t)) + 3e^{-3t}(\frac{1}{22}e^{4t} + \frac{1}{110}e^{3t}(\cos t - \sin t)) \\ -2e^{2t}(-\frac{5}{11}e^{-t} - \frac{3}{55}e^{-2t}(\cos t + 2\sin t)) + 5e^{-3t}(\frac{1}{22}e^{4t} + \frac{1}{110}e^{3t}(\cos t - 3\sin t)) \end{bmatrix}$$

$$\overrightarrow{x_p} = \begin{bmatrix} -\frac{5}{11}e^{t} - \frac{3}{55}(\cos t + 2\sin t) + \frac{3}{22}e^{t} + \frac{3}{110}(\cos t - \sin t) \\ \frac{10}{11}e^{t} + \frac{6}{55}(\cos t + 2\sin t) + \frac{5}{22}e^{t} + \frac{5}{110}(\cos t - 3\sin t) \end{bmatrix}$$

© Krista King Math

$$\overrightarrow{x_p} = \begin{bmatrix} -\frac{5}{11}e^t + \frac{3}{22}e^t - \frac{3}{55}\cos t + \frac{3}{110}\cos t - \frac{6}{55}\sin t - \frac{3}{110}\sin t \\ \frac{10}{11}e^t + \frac{5}{22}e^t + \frac{6}{55}\cos t + \frac{5}{110}\cos t + \frac{12}{55}\sin t - \frac{15}{110}\sin t \end{bmatrix}$$

$$\overrightarrow{x_p} = \begin{bmatrix} -\frac{7}{22}e^t - \frac{3}{110}\cos t - \frac{15}{110}\sin t \\ \frac{25}{22}e^t + \frac{17}{110}\cos t + \frac{9}{110}\sin t \end{bmatrix}$$

Breaking apart the particular solution gives

$$\overrightarrow{x_p} = \begin{bmatrix} -\frac{7}{22}e^t \\ \frac{25}{22}e^t \end{bmatrix} + \begin{bmatrix} -\frac{3}{110}\cos t \\ \frac{17}{110}\cos t \end{bmatrix} + \begin{bmatrix} -\frac{15}{110}\sin t \\ \frac{9}{110}\sin t \end{bmatrix}$$

$$\overrightarrow{x_p} = \frac{1}{22}\begin{bmatrix} -7 \\ 25 \end{bmatrix} e^t + \frac{1}{110}\begin{bmatrix} -3 \\ 17 \end{bmatrix}\cos t + \frac{1}{110}\begin{bmatrix} -15 \\ 9 \end{bmatrix}\sin t$$

© Krista King Math

Topic: Variation of parameters for nonhomogeneous systems

Question: Use the method of variation of parameters to find the general solution to the system.

$$x_1'(t) = 2x_1 + x_2$$

$$x_2'(t) = 3x_2 + te^t$$

Answer choices:

A $\vec{x} = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{3t} + \begin{bmatrix} 2 \\ -1 \end{bmatrix} te^t$

B $\vec{x} = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{3t} + \begin{bmatrix} 2 \\ -1 \end{bmatrix} te^t + \begin{bmatrix} 3 \\ -1 \end{bmatrix} e^t$

C $\vec{x} = c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{3t} + \frac{1}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix} te^t$

D $\vec{x} = c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{3t} + \frac{1}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix} te^t + \frac{1}{4} \begin{bmatrix} 3 \\ -1 \end{bmatrix} e^t$

© Krista King Math

Solution: D

Start by writing the nonhomogeneous system in matrix form.

$$\vec{x}' = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \vec{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} te^t$$

Now we'll find $|A - \lambda I|$.

$$|A - \lambda I| = \left| \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right|$$

$$|A - \lambda I| = \left| \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right|$$

$$|A - \lambda I| = \begin{vmatrix} 2 - \lambda & 1 \\ 0 & 3 - \lambda \end{vmatrix}$$

$$|A - \lambda I| = (2 - \lambda)(3 - \lambda) - (1)(0)$$

$$|A - \lambda I| = (2 - \lambda)(3 - \lambda)$$

So the characteristic equation is

$$(2 - \lambda)(3 - \lambda) = 0$$

$$\lambda = 2, \, 3$$

Then for these Eigenvalues, $\lambda_1 = 2$ and $\lambda_2 = 3$, we find

$$A - 2I = \begin{bmatrix} 2 - 2 & 1 \\ 0 & 3 - 2 \end{bmatrix}$$

© Krista King Math

$$A - 2I = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$$

and

$$A - 3I = \begin{bmatrix} 2-3 & 1 \\ 0 & 3-3 \end{bmatrix}$$

$$A - 3I = \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix}$$

Put these two matrices into reduced row-echelon form.

$$\begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} \qquad \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

If we turn these back into systems of equations, we get

$$k_2 = 0 \qquad -k_1 + k_2 = 0$$

$$k_1 = k_2$$

For the first system, we choose $(k_1, k_2) = (1,0)$. And from the second system, we choose $(k_1, k_2) = (1,1)$.

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad \overrightarrow{k_2} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Then the solutions to the system are

© Krista King Math

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t} \qquad \overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{2t} \qquad \overrightarrow{x_2} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{3t}$$

Therefore, the complementary solution to the nonhomogeneous system, which is also the general solution to the associated homogeneous system, will be

$$\overrightarrow{x_c} = c_1 \overrightarrow{x_1} + c_2 \overrightarrow{x_2}$$

$$\overrightarrow{x_c} = c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{3t}$$

Now we can turn to finding the particular solution. We'll use the solution vectors $\overrightarrow{x_1}$ and $\overrightarrow{x_2}$ to form $\Phi(t)$,

$$\Phi(t) = \begin{bmatrix} e^{2t} & e^{3t} \\ 0 & e^{3t} \end{bmatrix}$$

then we'll find its inverse $\Phi^{-1}(t)$ by changing $[\Phi(t)|I]$ into $[I|\Phi^{-1}(t)]$.

$$\left[\begin{array}{cc|cc} e^{2t} & e^{3t} & 1 & 0 \\ 0 & e^{3t} & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & e^{t} & e^{-2t} & 0 \\ 0 & e^{3t} & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & e^{t} & e^{-2t} & 0 \\ 0 & 1 & 0 & e^{-3t} \end{array}\right]$$

© Krista King Math

$$\left[\begin{array}{cc|cc} 1 & 0 & e^{-2t} & -e^{-2t} \\ 0 & 1 & 0 & e^{-3t} \end{array}\right]$$

Now with

$$\Phi(t) = \begin{bmatrix} e^{2t} & e^{3t} \\ 0 & e^{3t} \end{bmatrix} \qquad \Phi^{-1}(t) = \begin{bmatrix} e^{-2t} & -e^{-2t} \\ 0 & e^{-3t} \end{bmatrix} \qquad F(t) = \begin{bmatrix} 0 \\ te^{t} \end{bmatrix}$$

the particular solution from the method of variation of parameters is given by

$$\overrightarrow{x_p} = \Phi(t) \int \Phi^{-1}(t) F(t)\ dt$$

$$\overrightarrow{x_p} = \begin{bmatrix} e^{2t} & e^{3t} \\ 0 & e^{3t} \end{bmatrix} \int \begin{bmatrix} e^{-2t} & -e^{-2t} \\ 0 & e^{-3t} \end{bmatrix} \begin{bmatrix} 0 \\ te^{t} \end{bmatrix}\ dt$$

$$\overrightarrow{x_p} = \begin{bmatrix} e^{2t} & e^{3t} \\ 0 & e^{3t} \end{bmatrix} \int \begin{bmatrix} e^{-2t}(0) - e^{-2t}(te^{t}) \\ 0(0) + e^{-3t}(te^{t}) \end{bmatrix}\ dt$$

$$\overrightarrow{x_p} = \begin{bmatrix} e^{2t} & e^{3t} \\ 0 & e^{3t} \end{bmatrix} \int \begin{bmatrix} -te^{-t} \\ te^{-2t} \end{bmatrix}\ dt$$

Integrate, then simplify the result.

$$\overrightarrow{x_p} = \begin{bmatrix} e^{2t} & e^{3t} \\ 0 & e^{3t} \end{bmatrix} \begin{bmatrix} te^{-t} + e^{-t} \\ -\frac{1}{2}te^{-2t} - \frac{1}{4}e^{-2t} \end{bmatrix}$$

$$\overrightarrow{x_p} = \begin{bmatrix} e^{2t}(te^{-t} + e^{-t}) + e^{3t}\left(-\frac{1}{2}te^{-2t} - \frac{1}{4}e^{-2t}\right) \\ 0(te^{-t} + e^{-t}) + e^{3t}\left(-\frac{1}{2}te^{-2t} - \frac{1}{4}e^{-2t}\right) \end{bmatrix}$$

© Krista King Math

$$\overrightarrow{x_p} = \begin{bmatrix} \frac{1}{2}te^t + \frac{3}{4}e^t \\ -\frac{1}{2}te^t - \frac{1}{4}e^t \end{bmatrix}$$

Summing the complementary and particular solutions gives the general solution to the nonhomogeneous system of differential equations.

$$\overrightarrow{x} = \overrightarrow{x_c} + \overrightarrow{x_p}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{3t} + \begin{bmatrix} \frac{1}{2}te^t + \frac{3}{4}e^t \\ -\frac{1}{2}te^t - \frac{1}{4}e^t \end{bmatrix}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{3t} + \frac{1}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix} te^t + \frac{1}{4} \begin{bmatrix} 3 \\ -1 \end{bmatrix} e^t$$

© Krista King Math

Topic: Variation of parameters for nonhomogeneous systems

Question: Use the method of variation of parameters to find the general solution to the system.

$$\vec{x}' = \begin{bmatrix} 2 & -1 \\ 6 & -3 \end{bmatrix} \vec{x} + \begin{bmatrix} e^{2t} \\ e^t + 1 \end{bmatrix}$$

Answer choices:

A $\vec{x} = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{-t} + \begin{bmatrix} \frac{5}{6} \\ 1 \end{bmatrix} e^t$

B $\vec{x} = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{-t} + \begin{bmatrix} \frac{5}{6} \\ 1 \end{bmatrix} e^t - \begin{bmatrix} 1 \\ 2 \end{bmatrix} t + \begin{bmatrix} 1 \\ 3 \end{bmatrix}$

C $\vec{x} = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{-t} + \begin{bmatrix} \frac{5}{6} \\ 1 \end{bmatrix} e^{2t} - \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^t$

D $\vec{x} = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{-t} + \begin{bmatrix} \frac{5}{6} \\ 1 \end{bmatrix} e^{2t} - \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^t - \begin{bmatrix} 1 \\ 2 \end{bmatrix} t + \begin{bmatrix} 1 \\ 3 \end{bmatrix}$

© Krista King Math

Solution: D

Start by finding $|A - \lambda I|$.

$$|A - \lambda I| = \left| \begin{bmatrix} 2 & -1 \\ 6 & -3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right|$$

$$|A - \lambda I| = \left| \begin{bmatrix} 2 & -1 \\ 6 & -3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right|$$

$$|A - \lambda I| = \begin{vmatrix} 2 - \lambda & -1 \\ 6 & -3 - \lambda \end{vmatrix}$$

$$|A - \lambda I| = (2 - \lambda)(-3 - \lambda) - (-1)(6)$$

$$|A - \lambda I| = -6 + \lambda + \lambda^2 + 6$$

So the characteristic equation is

$$\lambda^2 + \lambda = 0$$

$$\lambda(\lambda + 1) = 0$$

$$\lambda = 0, \ -1$$

Then for these Eigenvalues we find

$$A - 0I = \begin{bmatrix} 2 - 0 & -1 \\ 6 & -3 - 0 \end{bmatrix}$$

$$A - 0I = \begin{bmatrix} 2 & -1 \\ 6 & -3 \end{bmatrix}$$

© Krista King Math

and

$$A-(-1)I=\begin{bmatrix}2-(-1) & -1\\ 6 & -3-(-1)\end{bmatrix}$$

$$A-(-1)I=\begin{bmatrix}3 & -1\\ 6 & -2\end{bmatrix}$$

Put these two matrices into reduced row-echelon form.

$$\begin{bmatrix}2 & -1\\ 6 & -3\end{bmatrix} \qquad \begin{bmatrix}3 & -1\\ 6 & -2\end{bmatrix}$$

$$\begin{bmatrix}1 & -\frac{1}{2}\\ 6 & -3\end{bmatrix} \qquad \begin{bmatrix}1 & -\frac{1}{3}\\ 6 & -2\end{bmatrix}$$

$$\begin{bmatrix}1 & -\frac{1}{2}\\ 0 & 0\end{bmatrix} \qquad \begin{bmatrix}1 & -\frac{1}{3}\\ 0 & 0\end{bmatrix}$$

If we turn these back into systems of equations, we get

$$k_1-\frac{1}{2}k_2=0 \qquad k_1-\frac{1}{3}k_2=0$$

$$k_1=\frac{1}{2}k_2 \qquad k_1=\frac{1}{3}k_2$$

For the first system, we choose $(k_1,k_2)=(1,2)$. And from the second system, we choose $(k_1,k_2)=(1,3)$.

$$\vec{k_1}=\begin{bmatrix}1\\2\end{bmatrix} \qquad \vec{k_2}=\begin{bmatrix}1\\3\end{bmatrix}$$

© Krista King Math

Then the solutions to the system are

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t} \qquad \overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \qquad \overrightarrow{x_2} = \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{-t}$$

Therefore, the complementary solution to the nonhomogeneous system, which is also the general solution to the associated homogeneous system, will be

$$\overrightarrow{x_c} = c_1 \overrightarrow{x_1} + c_2 \overrightarrow{x_2}$$

$$\overrightarrow{x_c} = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{-t}$$

Now we can turn to finding the particular solution. We'll use the solution vectors $\overrightarrow{x_1}$ and $\overrightarrow{x_2}$ to form $\Phi(t)$,

$$\Phi(t) = \begin{bmatrix} 1 & e^{-t} \\ 2 & 3e^{-t} \end{bmatrix}$$

then we'll find its inverse $\Phi^{-1}(t)$ by changing $[\Phi(t)\,|\,I]$ into $[I\,|\,\Phi^{-1}(t)]$.

$$\left[\begin{array}{cc|cc} 1 & e^{-t} & 1 & 0 \\ 2 & 3e^{-t} & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & e^{-t} & 1 & 0 \\ 0 & e^{-t} & -2 & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & e^{-t} & 1 & 0 \\ 0 & 1 & -2e^{t} & e^{t} \end{array}\right]$$

© Krista King Math

$$\left[\begin{array}{cc|cc} 1 & 0 & 3 & -1 \\ 0 & 1 & -2e^t & e^t \end{array}\right]$$

Now with

$$\Phi(t) = \begin{bmatrix} 1 & e^{-t} \\ 2 & 3e^{-t} \end{bmatrix} \qquad \Phi^{-1}(t) = \begin{bmatrix} 3 & -1 \\ -2e^t & e^t \end{bmatrix} \qquad F(t) = \begin{bmatrix} e^{2t} \\ e^t + 1 \end{bmatrix}$$

the particular solution from the method of variation of parameters is given by

$$\overrightarrow{x_p} = \Phi(t) \int \Phi^{-1}(t) F(t)\, dt$$

$$\overrightarrow{x_p} = \begin{bmatrix} 1 & e^{-t} \\ 2 & 3e^{-t} \end{bmatrix} \int \begin{bmatrix} 3 & -1 \\ -2e^t & e^t \end{bmatrix} \begin{bmatrix} e^{2t} \\ e^t + 1 \end{bmatrix} dt$$

$$\overrightarrow{x_p} = \begin{bmatrix} 1 & e^{-t} \\ 2 & 3e^{-t} \end{bmatrix} \int \begin{bmatrix} 3(e^{2t}) - 1(e^t + 1) \\ -2e^t(e^{2t}) + e^t(e^t + 1) \end{bmatrix} dt$$

$$\overrightarrow{x_p} = \begin{bmatrix} 1 & e^{-t} \\ 2 & 3e^{-t} \end{bmatrix} \int \begin{bmatrix} 3e^{2t} - e^t - 1 \\ -2e^{3t} + e^{2t} + e^t \end{bmatrix} dt$$

Integrate, then simplify the result.

$$\overrightarrow{x_p} = \begin{bmatrix} 1 & e^{-t} \\ 2 & 3e^{-t} \end{bmatrix} \begin{bmatrix} \frac{3}{2}e^{2t} - e^t - t \\ -\frac{2}{3}e^{3t} + \frac{1}{2}e^{2t} + e^t \end{bmatrix}$$

$$\overrightarrow{x_p} = \begin{bmatrix} 1\left(\frac{3}{2}e^{2t} - e^t - t\right) + e^{-t}\left(-\frac{2}{3}e^{3t} + \frac{1}{2}e^{2t} + e^t\right) \\ 2\left(\frac{3}{2}e^{2t} - e^t - t\right) + 3e^{-t}\left(-\frac{2}{3}e^{3t} + \frac{1}{2}e^{2t} + e^t\right) \end{bmatrix}$$

© Krista King Math

$$\overrightarrow{x_p} = \begin{bmatrix} \frac{5}{6}e^{2t} - \frac{1}{2}e^{t} - t + 1 \\ e^{2t} - \frac{1}{2}e^{t} - 2t + 3 \end{bmatrix}$$

Summing the complementary and particular solutions gives the general solution to the nonhomogeneous system of differential equations.

$$\overrightarrow{x} = \overrightarrow{x_c} + \overrightarrow{x_p}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{-t} + \begin{bmatrix} \frac{5}{6}e^{2t} - \frac{1}{2}e^{t} - t + 1 \\ e^{2t} - \frac{1}{2}e^{t} - 2t + 3 \end{bmatrix}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{-t} + \begin{bmatrix} \frac{5}{6} \\ 1 \end{bmatrix} e^{2t} - \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{t} - \begin{bmatrix} 1 \\ 2 \end{bmatrix} t + \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

© Krista King Math